

TOPIC 2.4

How do atoms rearrange in different types of chemical reactions?

Key Concepts

- A compound forms in a synthesis reaction and breaks down in a decomposition reaction.
- In replacement reactions, elements replace other elements.
- Most combustion reactions release heat and light.
- In a neutralization reaction, an acid reacts with a base.

Curricular Competencies

- Formulate multiple hypotheses and predict multiple outcomes.
- Select and use appropriate equipment to systematically and accurately collect and record data.
- Analyze cause-and-effect relationships.
- Evaluate the validity and limitations of a model or analogy.

British Columbia is home to numerous breath-taking and distinctive landscapes, including karst formations that run deep beneath our feet. These are the many caves, sinkholes, and complex underground networks of springs and streams—all the result of chemical reactions between certain kinds of rock, such as limestone, and rainwater.

Carbon dioxide in the atmosphere dissolves in rainwater. A reaction between water and carbon dioxide produces a compound called carbonic acid, $\text{H}_2\text{CO}_3(\text{aq})$. As rainwater falls and infiltrates the land over thousands of years, carbonic acid reacts with compounds in bedrock such as limestone, slowly dissolving the rock and carving out karst terrain. Each year, spelunkers (cave explorers) from the province, country, and around the world come to study and to marvel at these examples of nature.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Identifying Preconceptions** What do you know about acids and bases? Share your ideas with a partner or in small groups.
2. **Processing and Analyzing** In what parts of B.C. would you expect to find karst formations?
3. **Predicting** Considering what you know about elements and compounds, what could happen when
 - an element interacts with an ionic compound
 - two ionic compounds interact
 - a compound breaks apart
 - two elements combine
4. **Considering First Peoples' Perspectives** Karst caves are often fragile and may be destroyed during resource extraction such as mining unless they have archeological or spiritual importance. Why might First Nations communities want to protect karst caves in their territories?



Key Terms

There are 10 key terms that are highlighted in bold type in this topic:

- | | |
|-------------------------------|-------------------------------|
| • synthesis reaction | • decomposition reaction |
| • single replacement reaction | • double replacement reaction |
| • combustion reaction | • acid |
| • base | • acid-base indicator |
| • pH scale | • neutralization reaction |

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meaning. Add other terms that you think are important and want to remember.

CONCEPT 1

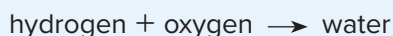
A compound forms in a synthesis reaction and breaks down in a decomposition reaction.

Activity

Recognizing Patterns: Building Your Own Summary



Make simple drawings that represent the atoms and bonds in the reactions below. Write the balanced chemical equations. Compare the two reactions. Describe what bonds are broken, what bonds form, and how atoms are rearranged.



As you complete this concept, identify the type of reaction each represents. As you study this Topic, add more information to this material. Come up with models for each of the reaction types using examples other than chemical symbols. What do the reactants typically look like in each type of reaction? What do the products typically look like in each type of reaction?

synthesis reaction

a chemical reaction in which two or more reactants combine to produce a single product

Synthesis Reactions

In a **synthesis reaction**, two or more elements or simple compounds combine to form a new compound. The reactants can be any combination of elements or compounds. However, the product is always a single compound. Most synthesis reactions are exothermic.

The general form of a synthesis reaction is shown below in two ways. The first way uses the individual letters A and B to represent the components. The second way uses coloured circles.

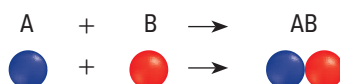


Figure 2.24 shows a few of the many examples of synthesis reactions in the world around you. **Figure 2.25** uses images to show both the visible, large-scale view and the atomic-scale view of a synthesis reaction.

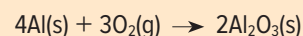
Figure 2.24 Examples of common synthesis reactions.



Dull Aluminum

Shiny aluminum rims would soon become dull after they are installed on a car if they were not protected by a clear coat. The dull sheen that forms on unprotected aluminum is aluminum oxide. Aluminum oxide is seen on any unprotected aluminum that is exposed to air, including boats and window frames.

aluminum + oxygen $\rightarrow$ aluminum oxide



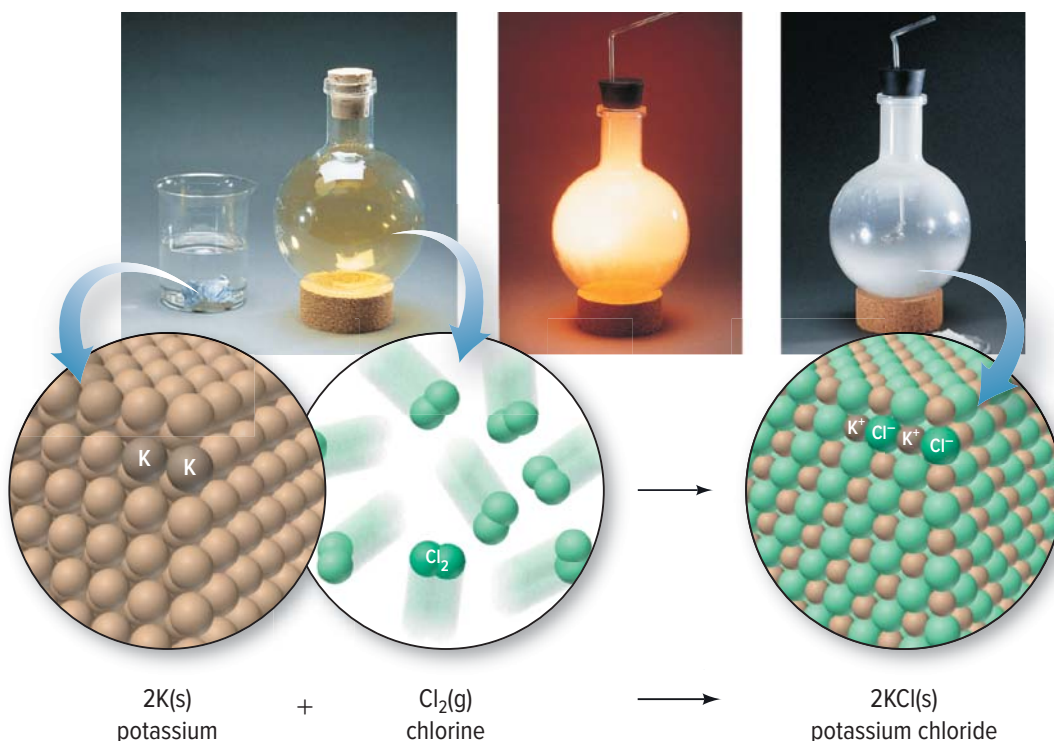
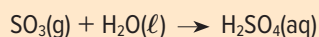
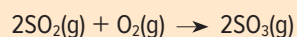


Figure 2.25 Potassium and chlorine combine in a dramatic synthesis reaction to form potassium chloride. **Analyzing:** Is this an exothermic or endothermic reaction? Provide evidence to support your answer.



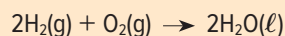
Acid Precipitation

Industrial processes and burning coal with sulfur impurities produce sulfur compounds. These combine with moisture in the air to produce acid precipitation (acidic rain, snow, and fog). Acid precipitation damages buildings and harms plants and aquatic life. There are two different synthesis reactions that contribute to acid precipitation.



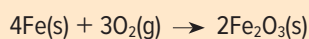
Hydrogen-Powered Vehicles

In hydrogen fuel cells, hydrogen and oxygen are combined to produce water, which is the only chemical product. The energy produced by this reaction is converted to electricity that is used to power vehicles.



Metal Rusting

Damage to equipment, buildings, and structures caused by metal corrosion costs society billions of dollars each year. Rusting is a type of corrosion that involves the reaction of iron. The process is complex and occurs in multiple steps. The final product is iron(III) oxide, Fe₂O₃(s), which has a brownish-orange colour. A general chemical equation that represents rust formation is



decomposition reaction

a chemical reaction in which a compound is broken down into elements or simpler compounds

Decomposition Reactions

In a **decomposition reaction**, one compound breaks down into two or more elements or simpler compounds. A decomposition reaction occurs when a reactant absorbs enough energy for one or more of its bonds to break. The energy can be in different forms—heat and electricity are two common ones. A decomposition reaction is often the reverse of a synthesis reaction. Therefore, most decomposition reactions are endothermic. The general form of a decomposition reaction is shown here.

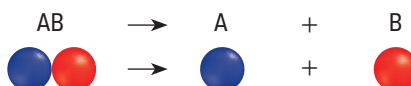
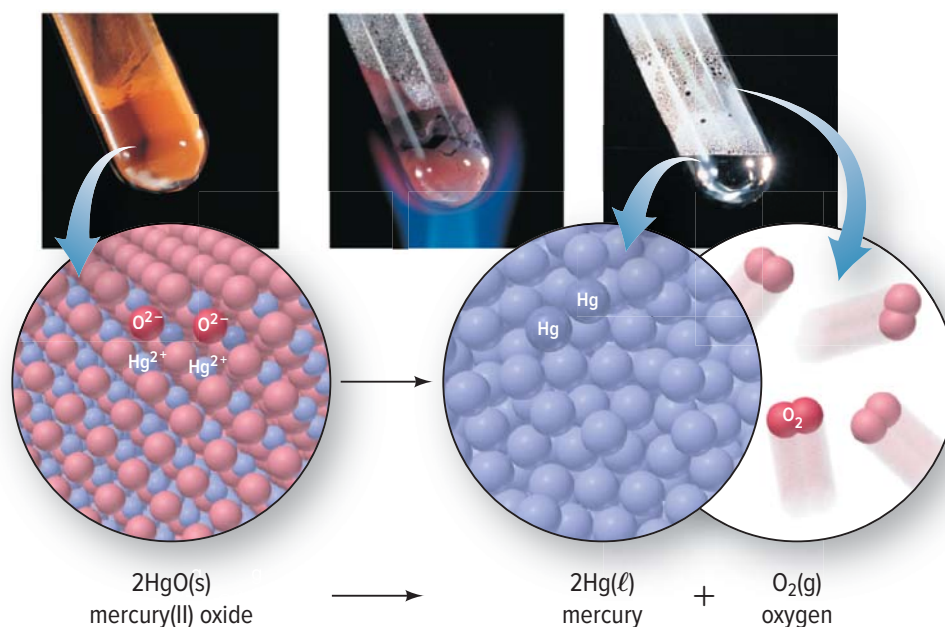


Figure 2.26 shows what happens when mercury(II) oxide decomposes. This compound is used in mercury batteries.

Figure 2.26 Solid mercury(II) oxide breaks down in a decomposition reaction. **Communicating:** Describe the overall energy change in this reaction.



Activity

What Happened to the Baking Soda?

The chemical equation for the reaction in this demonstration is



1. Your teacher will set up a system made up of a test tube, test tube clamp, ring stand, 2 g baking soda, and Bunsen burner. Record the mass of the test tube and baking soda your teacher provides.
2. Observe as the baking soda and test tube are heated gently and slowly using a cool flame for about 2 min. Record your observations.
3. Your teacher will measure the mass of the test tube and material after heating. Record it.
4. Compare the mass before and after heating. Account for any differences.
5. Was this system open or closed? Explain.

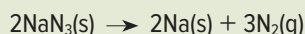
Figure 2.27 shows a few of the many decomposition reactions in the world around you.



Motorcycle Air Bags

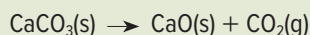
Many motorcycles have air bags that inflate within thousandths of a second in a crash. A motorcycle changes speed very quickly during a crash. This triggers a sensor in the air bag to create an electric impulse. The impulse sparks sodium azide in the air bag. Sodium azide immediately breaks down into solid sodium and nitrogen gas. The nitrogen gas fills the bag to cushion the driver on impact.

The decomposition reaction is



Limestone and Cement

Limestone is a raw material in a host of products used in construction, agriculture, and the pulp and paper industry. The traditional territory of the Lheidli T'Enneh First Nation includes significant limestone deposits that may be quarried. Limestone is calcium carbonate, which yields calcium oxide (quick lime) and carbon dioxide when heated, as shown in the chemical equation below. The calcium oxide produced in this decomposition reaction is the main component of cement. Cement is added to water, sand, and gravel to make concrete. The calcium oxide acts as a binding agent. It holds the mixture together as the concrete dries and cures over time.



Explosion for Demolition Involving TNT

Many explosives produce tremendous force to break apart materials. They can occur as a result of the decomposition of a compound and the rapid heating and expansion of the gases that are produced. For example, trinitrotoluene (TNT) is a commonly used explosive that decomposes into elements and compounds.

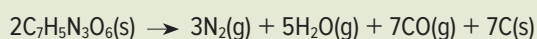


Figure 2.27 The energy used to initiate decomposition reactions may come from heat, electrical energy, and light. **Applying:** How many examples of reactions involving these three energy inputs can you discover?

Before you leave this page . . .

1. In your own words, describe what happens in a synthesis reaction and a decomposition reaction.
2. Which of the following is a synthesis reaction and which is a decomposition reaction? Predict the products formed in each reaction, and give the balanced chemical equations.
 - a) $\text{Al}(\text{s}) + \text{F}_2(\text{g}) \rightarrow$
 - b) $\text{AgCl}(\text{s}) \rightarrow$
3. Are decomposition reactions endothermic or exothermic? Justify your response through a discussion of the energy changes associated with breaking and forming bonds.

CONCEPT 2

In replacement reactions, elements replace other elements.

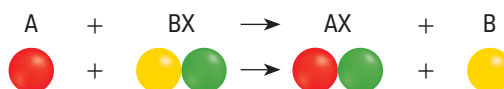
single replacement reaction a chemical reaction in which an element and a compound react to produce another element and another compound

Single Replacement Reactions

In a **single replacement reaction**, an element and a compound react to produce another element and another compound. In this type of reaction, an element takes the place of another element in a compound.

A single replacement reaction has two general forms. These are shown below. The first equation represents a metal element (A) replacing the metal ion (B) in a compound. The second equation represents a non-metal element (Y) replacing the non-metal ion in a compound (X).

Reactions in which a metal replaces another metal:



Reactions in which a non-metal replaces another non-metal:



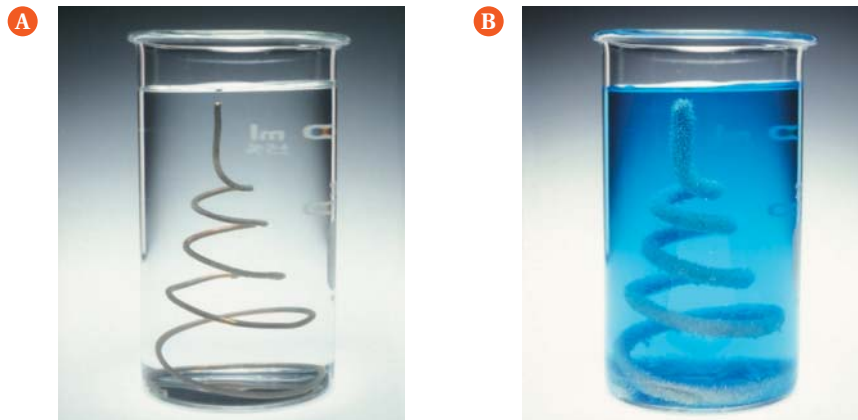
A Metal Can Replace a Metal

Figure 2.28 shows the following single replacement reaction, which involves a metal atom replacing a metal ion in a compound.



In this example, the metal element copper replaces the silver ion in the compound silver nitrate. Usually, the compound reacting must be dissolved in water to form an aqueous solution. The new compound that forms is copper(II) nitrate, which is dissolved in the solution, and silver now exists as solid crystals on the wire.

Figure 2.28 In **A**, a coil of copper wire is placed in a solution of silver nitrate. The fuzzy coating on the copper wire in **B** is silver metal. The blue colour of the solution is characteristic of Cu^{2+} ions in water.



Hydrogen Can Act as a Metal

In a single replacement reaction, hydrogen in water can behave as a metal.

Figure 2.29 shows how lithium replaces hydrogen in water.

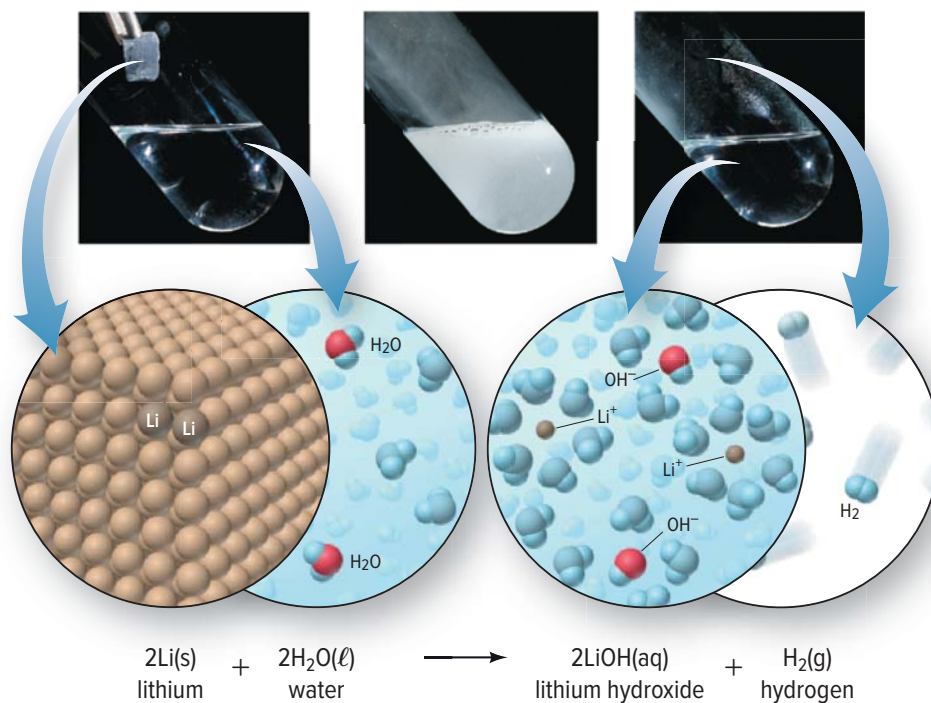


Figure 2.29 Lithium replaces hydrogen in a single replacement reaction. The bubbles in the test tube are hydrogen gas that forms in the reaction.

A Non-metal Can Replace a Non-metal

An example of the second type of single replacement reaction is the reaction of chlorine gas with a solution of sodium bromide, shown in **Figure 2.30**.

In this reaction, chlorine replaces bromide ions in the solution, resulting in the formation of sodium chloride and bromine. **Figure 2.31** on the next page shows a few examples of applications of single replacement reactions.



Figure 2.30 Chlorine gas (on the left) is bubbled through an aqueous solution of sodium bromide (on the right). The brown colour is the bromine produced.

Applying: Is this an open or closed system? Why do you think this system was used?



Welding Railroad Tracks

Some single replacement reactions release a lot of energy. In the railroad industry, for example, powdered aluminum and iron(III) oxide are mixed together. They react to form iron and aluminum oxide. When aluminum replaces the iron in iron(III) oxide, enough heat is released to melt the iron that is produced and weld the railroad tracks together.



Fire Retardants

More than half of the bromine produced on Earth is used in flame retardants in mattresses, furniture, and electronics. They make it harder for these items to catch fire and slow the spread of fire so people can escape their homes more easily. Bromine is produced commercially in the single replacement reaction shown in **Figure 2.30**.



Joining Electrical Conductors

Single replacement reactions are used to produce pure copper. This metal is used to join electrical conductors. Aluminum replaces copper in copper(II) oxide. The heat resulting from the reaction melts the copper that forms. The copper flows into a mold around the ends of the conductors. When the metal cools, it conducts electricity between the conductors.



Figure 2.31 There is a wide range of uses for single replacement reactions.

Extending the Connections

Mercurial Mysteries in the Arctic

Something strange has been happening in the circumpolar Arctic region. There is steam rising from cracks in Arctic sea ice cover during summer months. This steam is part of the answer to a mystery surrounding the apparent disappearance of mercury from the atmosphere during the Arctic summer. The solution to the mystery has involved scientists from Canada as well as an investigation led by NASA called BROMEX, which stands for Bromine, Ozone, and Mercury Experiment. But this mystery is more than just a matter of scientific curiosity. At stake are the lives, lifestyles, and livelihoods of Arctic First Peoples. What connections can you discover related to mercury, bromine, and single replacement reactions? Why is there more mercury in the Arctic than anywhere else on Earth? And what are the implications of all this mercury for the people, other animals, and the land of the Arctic?



Double Replacement Reactions

In a **double replacement reaction**, the positive ions of two ionic compounds change places to form two new ionic compounds. Double replacement reactions usually occur between ionic compounds in aqueous solution, meaning they are both dissolved in water. Double replacement reactions can be represented by a general chemical equation and pictorially, as shown below. In all double replacement reactions, A and B are positively charged ions. X and Y are negatively charged ions. Notice how the ions in the reactants are exchanged to form the products.

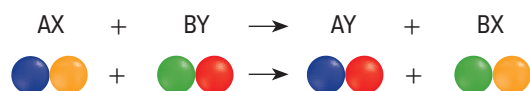


Figure 2.32 shows a double replacement reaction between aqueous solutions of potassium iodide, $KI(aq)$, and lead(II) nitrate, $Pb(NO_3)_2(aq)$. Notice the formation of a bright yellow solid. The formation of a solid, called a precipitate, is common in double replacement reactions.

You or someone in your family may have had first-hand experience with the products of double replacement reactions, such as those shown in **Figure 2.33**.

double replacement reaction a chemical reaction in which solutions of two ionic compounds react to produce two new compounds

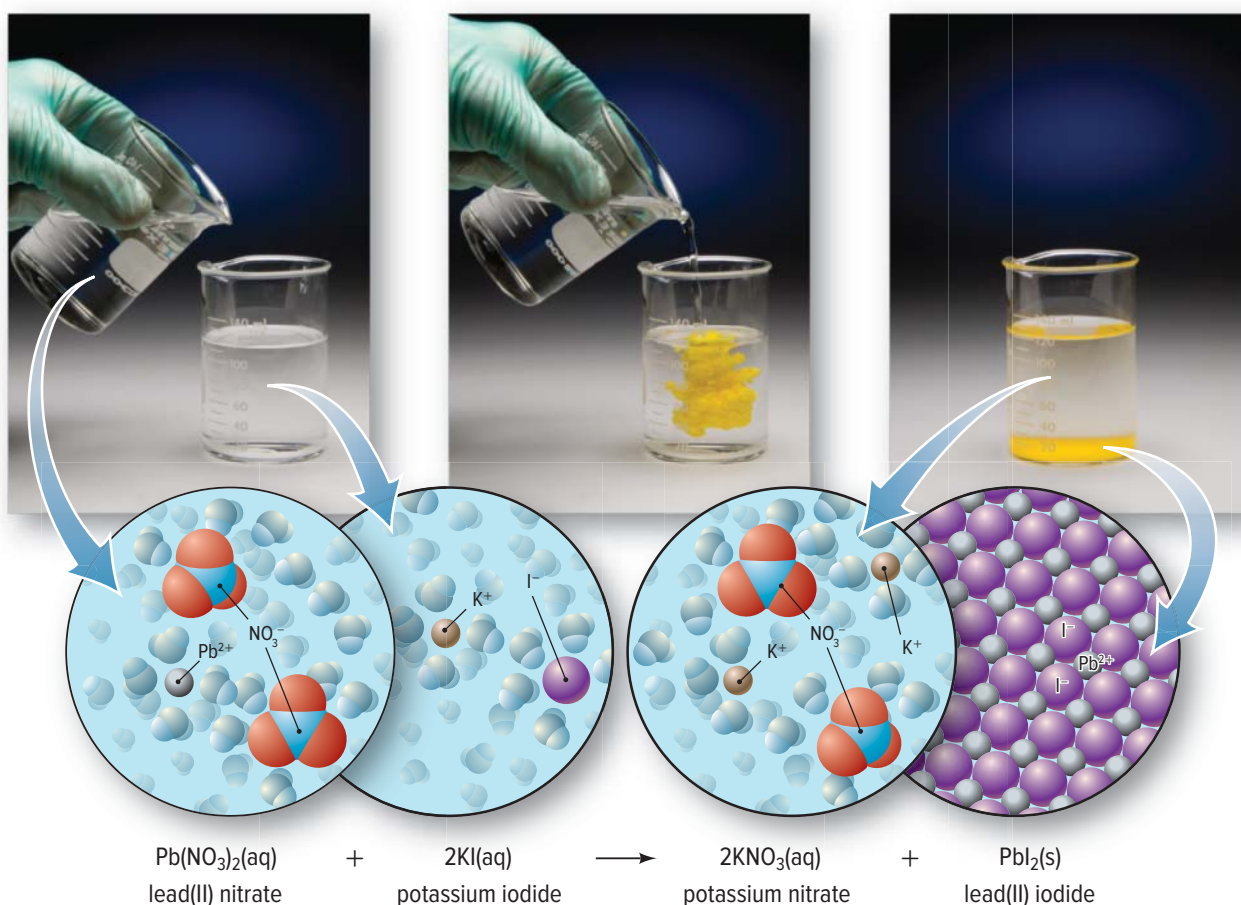
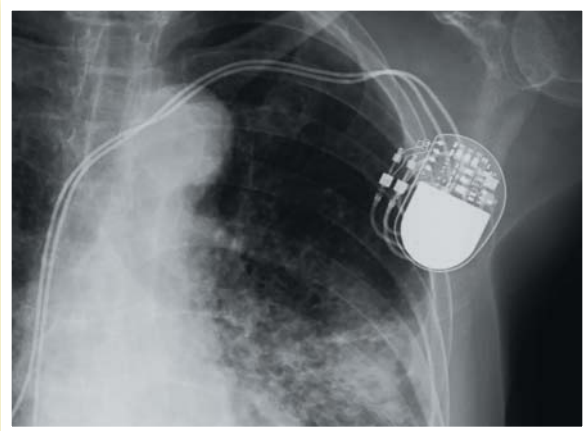
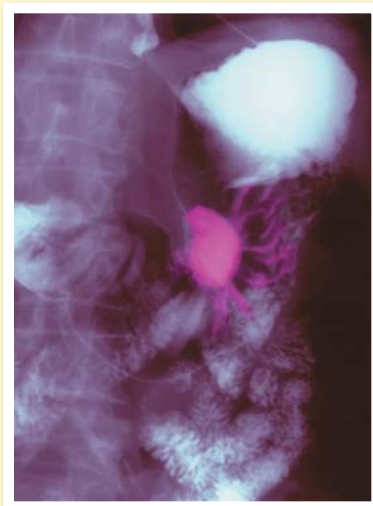
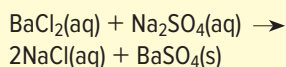


Figure 2.32 A yellow solid forms when potassium iodide and lead(II) nitrate undergo a double replacement reaction.

Soft Tissue X Rays

X rays are great for diagnosing broken bones. However, they do a poor job of showing soft tissue like the stomach, because X rays pass right through soft tissue. X-ray technicians get around this problem by having patients drink a chalky liquid that contains solid barium sulfate. This ionic compound blocks X rays. As a result, the soft tissue can be seen in the X-ray image. Barium sulfate is produced in a double replacement reaction.



Pacemaker Batteries

Do you know someone who has a medical device inside their body that runs on batteries? Lithium batteries containing silver chromate, $\text{Ag}_2\text{CrO}_4(\text{s})$, are used in pacemakers. These devices help maintain the steady beat of the human heart. The silver chromate in these batteries is produced in a double replacement reaction. Solutions of silver nitrate and sodium chromate react to produce solid silver chromate and a solution of sodium nitrate.

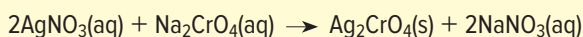


Figure 2.33 Two examples of double replacement reactions. **Questioning:** Examine the state symbols in the chemical equations—what questions do you have about these types of reactions?

Activity

Mix and Switch



1. Label two test tubes A and B. To test tube A, add a scoop of solid copper(II) sulfate, CuSO_4 . Add enough to just fill the rounded end of the test tube. To test tube B, add a similar amount of calcium chloride, CaCl_2 .
2. Add 10 mL of water to each test tube. Stopper each test tube and gently shake them to dissolve the solids. Record your observations.
3. Coil 10 cm of magnesium ribbon around a pencil. Place the coil in test tube A. Record your observations over 10 min.
4. Use a pipette to transfer 5 mL of liquid from test tube A to test tube B. Record your observations over 10 min.
5. Write balanced chemical equations for the reactions. Identify the types of reactions.



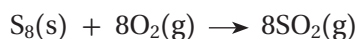
Before you leave this page . . .

1. Give the predicted products and the balanced chemical equations.
 - a) $\text{Mg}(\text{s}) + \text{AgCl}(\text{aq}) \rightarrow$
 - b) $\text{AgNO}_3(\text{aq}) + \text{K}_2\text{CrO}_4(\text{aq}) \rightarrow$
2. Identify each reaction in question 1 as single or double replacement.
3. In your own words, describe what happens in a single replacement reaction.

CONCEPT 3

Most combustion reactions release heat and light.

A **combustion reaction** is a chemical reaction in which a compound or element reacts with oxygen to form compounds called oxides. The oxides contain oxygen and the elements that make up the reactant compound. For example, sulfur (or compounds containing sulfur) will react with oxygen in a combustion reaction to produce sulfur dioxide, $\text{SO}_2(\text{g})$.



This reaction is one of the main sources of air pollution, as shown in **Figure 2.34**. The diagram also shows the impact of a similar reaction that involves nitrogen and oxygen. These products, sulfur dioxide and nitrogen dioxide, undergo a synthesis reaction with water to produce a type of compound called an acid. You will investigate acids and their impact on the environment in Concept 4.

Another feature of combustion reactions is that many produce heat and light. The burning of fuels such as natural gas and gasoline involves the combustion of a type of compound called a hydrocarbon. Hydrocarbons are compounds made up of just two elements: carbon and hydrogen. A general equation for the combustion of hydrocarbons is



In this general equation, C_xH_y is a general formula for hydrocarbons, with x representing the number of carbon atoms and y representing the number of hydrogen atoms that make up the compound. **Figure 2.35** shows some examples of common combustion reactions.

combustion reaction a chemical reaction in which an element reacts with oxygen to produce an oxide of the element and heat; also refers to the burning of hydrocarbons to produce carbon dioxide and water

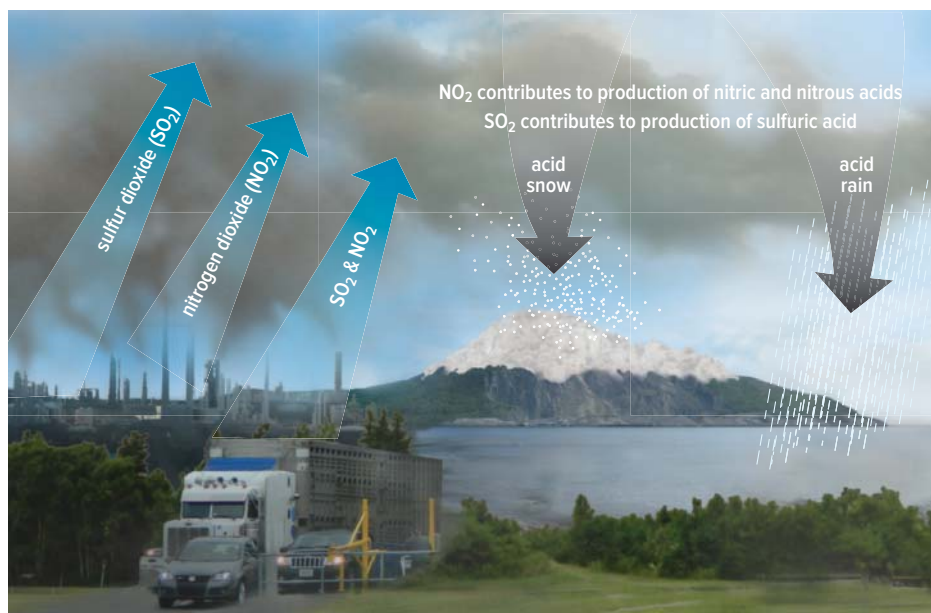


Figure 2.34 Key sources of SO_2 and NO_2 . **Questioning:** What sources of these pollutants are identified in this Figure? What effects do they have on human health and the environment?



Gasoline Combustion

Gasoline is a mixture of different hydrocarbons. Fuels are graded based on their octane rating. The greater the amount of octane present, the higher the quality of fuel. The combustion of octane is represented by the equation



Methane Combustion

Methane, $\text{CH}_4(\text{g})$, is a hydrocarbon that is the main component in natural gas. The flame and heat produced when using a gas stove is due to the combustion of methane. The balanced chemical equation for the complete combustion of methane is



Propane Combustion

Propane, $\text{C}_3\text{H}_8(\text{g})$, is a common fuel with several applications. Small propane torches like this one are often used by professionals, such as plumbers, and many people have them in their homes. Propane is also used as a heating fuel, with both indoor and outdoor heaters available. You are probably most familiar with its use as fuel for barbecues. Regardless of what the combustion of propane is used for, its combustion is represented by the equation



Figure 2.35 Our society depends on combustion reactions for transportation, cooking, and heating **Analyzing:** Are these combustion reactions exothermic or endothermic? What evidence supports your answer?

Extending the Connections

Combustion and Greenhouse Gases

The greenhouse effect is a natural part of Earth's climate system and helps to regulate Earth's temperature. Which greenhouse gases occur naturally, and which are the result of human activity? What role does combustion play in the greenhouse effect? Develop three questions that you have about this topic, and decide how you could investigate answers to them.

Figure 2.36 These are places where carbon monoxide produced from the incomplete combustion of fuels can occur. Accumulation of carbon monoxide in the air can be hazardous. **Applying:** Identify sources in your home where carbon monoxide could be produced.



Incomplete Combustion

If the supply of oxygen is too low, incomplete combustion occurs. Like complete combustion, incomplete combustion produces carbon dioxide and water. But it also results in other products such as carbon (soot) and carbon monoxide. Carbon monoxide is a colourless, odourless, highly toxic gas. You have probably heard or read warnings about the dangers associated with using camp stoves or propane barbecues indoors, or ever letting a car run in a closed garage. A poorly ventilated fireplace can also be dangerous. In fact, the burning of any fuels, such as gasoline, natural gas, oil, or wood, within a home or garage is potentially dangerous (**Figure 2.36**). If there is not enough oxygen, incomplete combustion results and carbon monoxide is produced. That's why carbon monoxide detectors like the one shown in **Figure 2.37** are necessary.



Figure 2.37 Homes with appliances or other equipment that burn fossil fuels and/or wood should have carbon monoxide detectors.

Activity

Considering Combustion

Methane is the hydrocarbon in natural gas used for school Bunsen burners. The photo shows two sets of burner flames. One set shows complete combustion; the other shows incomplete combustion. Which is which? What could explain the differences you observe in the flames? Share your ideas and reasoning with your teacher, who will decide how they can be explored safely in the lab or with video demonstrations.



Before you leave this page . . .

1. What energy changes are associated with hydrocarbon combustion?
2. What are the products of the complete combustion of a hydrocarbon?

CONCEPT 4

In a neutralization reaction, an acid reacts with a base.

Activity

What Reaction Type Is Neutralization?

Complete your summary of reaction types when you finish this concept. What are some ways that have helped you remember what happens to the atoms in the different types of reactions? After learning about neutralization reactions, describe another way you might classify this type of reaction.



Figure 2.38 Acids in citrus fruits such as this fingered citron makes them taste sour.

Many substances that you encounter every day are acids or bases. For example, the sour taste of citrus fruits such as grapefruit and lemons is due to the presence of acidic compounds (**Figure 2.38**). The bitter taste of coffee and the slippery feel of soap are due to the presence of basic compounds. Compounds may be classified as acids or bases depending on how they interact with water.

Acids

An **acid** is a substance that releases hydrogen ions, $\text{H}^+(\text{aq})$, in an aqueous solution. For example, when hydrogen chloride gas is mixed with water, $\text{H}^+(\text{aq})$ and $\text{Cl}^-(\text{aq})$ form. Therefore, $\text{HCl}(\text{aq})$ is called an acidic solution.



Acids include two broad categories. *Binary acids* are composed of two elements: hydrogen and a non-metal, such as $\text{HCl}(\text{aq})$. *Oxyacids* are composed of hydrogen, oxygen, and another element—for example, an acid with a polyatomic ion, such as $\text{H}_2\text{CO}_3(\text{aq})$. The chemical formula of an acid can often be recognized by the presence of one or more hydrogen ions with a negative ion. **Table 2.1** lists some common acids.

acid compound that forms H^+ ions when dissolved in water

Table 2.1 Common Acids and Their Uses

Chemical Formula	Classical Name	IUPAC Name	Uses
$\text{HCl}(\text{aq})$	hydrochloric acid	aqueous hydrogen chloride	producing plastics and processing metals
$\text{HF}(\text{aq})$	hydrofluoric acid	aqueous hydrogen fluoride	manufacturing and etching glass
$\text{H}_2\text{SO}_4(\text{aq})$	sulfuric acid	aqueous hydrogen sulfate	in car batteries
$\text{HNO}_3(\text{aq})$	nitric acid	aqueous hydrogen nitrate	making explosives and fertilizers
$\text{H}_2\text{CO}_3(\text{aq})$	carbonic acid	aqueous hydrogen carbonate	in carbonated drinks
$\text{CH}_3\text{COOH}(\text{aq})$	acetic acid	aqueous hydrogen acetate, or ethanoic acid	in vinegar

Bases

A **base** is a substance that releases hydroxide ions, $\text{OH}^-(\text{aq})$, in an aqueous solution. For example, when sodium hydroxide crystals dissolve in water, the ions $\text{Na}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$ form.



Therefore, $\text{NaOH}(\text{aq})$ is called a basic solution.

Table 2.2 lists the names, chemical formulas, and uses for some common bases. Many bases are ionic compounds composed of metal ions and hydroxide ions. The chemical formula for a base includes enough hydroxide ions in the formula to make the total charge of the compound zero.

base a compound that forms OH^- ions when dissolved in water

Table 2.2 Common Bases and Their Uses

Chemical Formula	Common Name	IUPAC Name	Uses
$\text{NaOH}(\text{aq})$	lye, caustic soda	sodium hydroxide	in drain cleaners, used in making soaps and paper
$\text{Mg}(\text{OH})_2(\text{aq})$	Milk of Magnesia®	magnesium hydroxide	antacids and laxatives
$\text{Ca}(\text{OH})_2(\text{aq})$	lime water	calcium hydroxide	soil and water treatment

Extending the Connections

Naming Acids and Bases

Like other compounds, acids and bases are named according to certain guidelines. Find out how the naming of acids and bases compares with naming other compounds.

Identifying Acids and Bases

Most common acids and bases form colourless solutions in water. One way to identify them is to use an **acid-base indicator**. This is a chemical that changes colour in response to whether it is exposed to an acid or a base. A common acid-base indicator is litmus. Litmus solution is often dried on thin paper strips (**Figure 2.39**).

acid-base indicator

a chemical that changes colour in response to the concentration of hydrogen ions in a solution

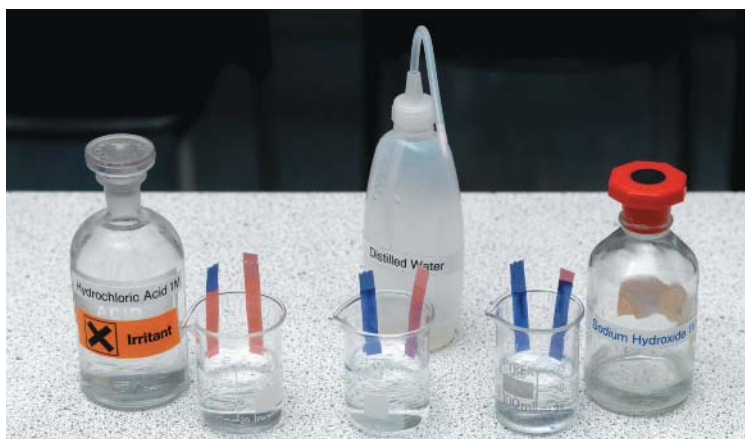


Figure 2.39 Blue litmus paper turns red in an acidic solution, and red litmus paper turns blue in a basic solution. **Evaluating:** Why did neither change colour in the water?

Activity

The Litmus Test



Your teacher will supply you with samples of materials to test. Make a prediction for each. Which samples do you think are basic, acidic, or neither? Place a small amount of each sample in a well of a spot plate. Make sure to keep track of what sample was added to which well. Dip a small piece of blue and red litmus paper into each well. Record your observations. How did your results compare with your predictions?

pH scale a numbered scale between 0 and 14 that indicates the acidity or basicity of a solution

Connect to Investigation 2-G on page 178

The pH Scale

To more accurately determine how acidic or basic a solution is, chemists use the pH scale and instruments to measure pH. The **pH scale** typically ranges from 0 to 14. It provides a way to measure the acidity and basicity of solutions. **Figure 2.40** shows the pH scale and pH values of some common substances.

Acidic Solutions: $\text{pH} < 7$

Acids have pH values below 7. The lower the pH value, the more acidic the solution. This means there are many more hydrogen ions in the solution than hydroxide ions. For example, a lemon at $\text{pH} = 2$ is more acidic than milk at $\text{pH} = 6$ and has a greater number of H^+ ions. Each time the pH value changes by one unit, the number of H^+ ions changes by 10 times. The lemon at $\text{pH} = 2$ has 10^4 or 10 000 times more hydrogen ions than milk at $\text{pH} = 6$.

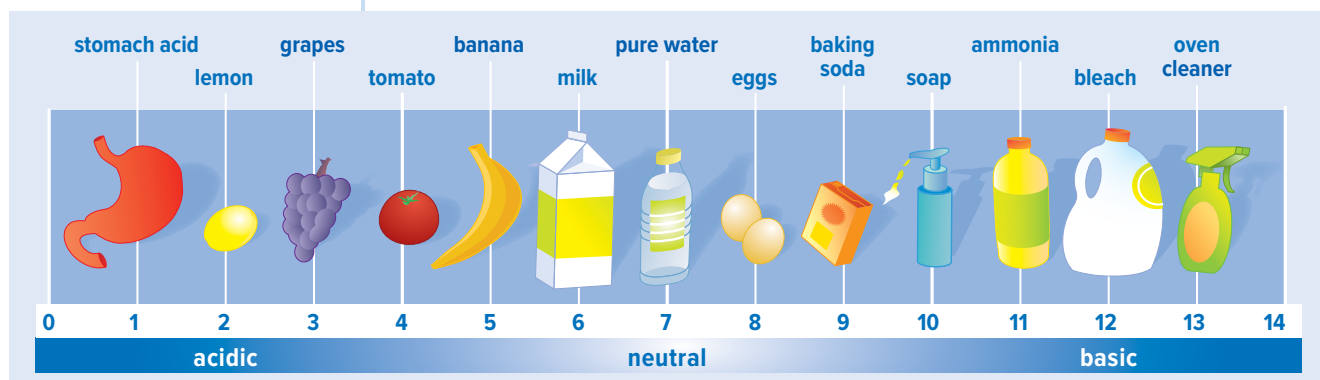
Basic Solutions: $\text{pH} > 7$

Bases have pH values above 7. This means there are many more hydroxide ions in the solution than hydrogen ions. The higher the pH, the more basic the solution is. For example, oven cleaner at $\text{pH} = 13$ is more basic than eggs at $\text{pH} = 8$.

Neutral Solutions: $\text{pH} = 7$

A solution that is neither acidic nor basic is neutral and falls in the middle of the pH scale at $\text{pH} = 7$. This means there are equal numbers of hydrogen ions and hydroxide ions in the solution. Pure water has a pH of 7, as do solutions of some compounds such as sodium chloride.

Figure 2.40 The pH scale ranges from 0 to 14.



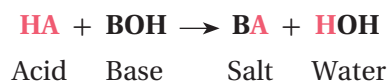
Extending the Connections

Measuring pH

In the laboratory, there are several ways to determine how acidic or basic a solution is. Some indicators can provide a range of possible pH values for the solution, while universal indicators turn different colours at particular pH values. The most accurate method is to use a pH meter, which directly measures the pH. Choose one of these approaches to investigate further. What questions do you have? How can you answer them?

Acid-Base Neutralization

A **neutralization reaction** is a chemical reaction between an acid and a base that produces water and a salt (a type of ionic compound). A general equation for the reaction between an acid and a base can be represented as



Notice that this reaction is actually a type of double replacement reaction. The ions of the reactants switch places to form two new compounds. The water forms as the hydrogen ions of the acid and the hydroxide ions of the base combine. The other ions form the salt.

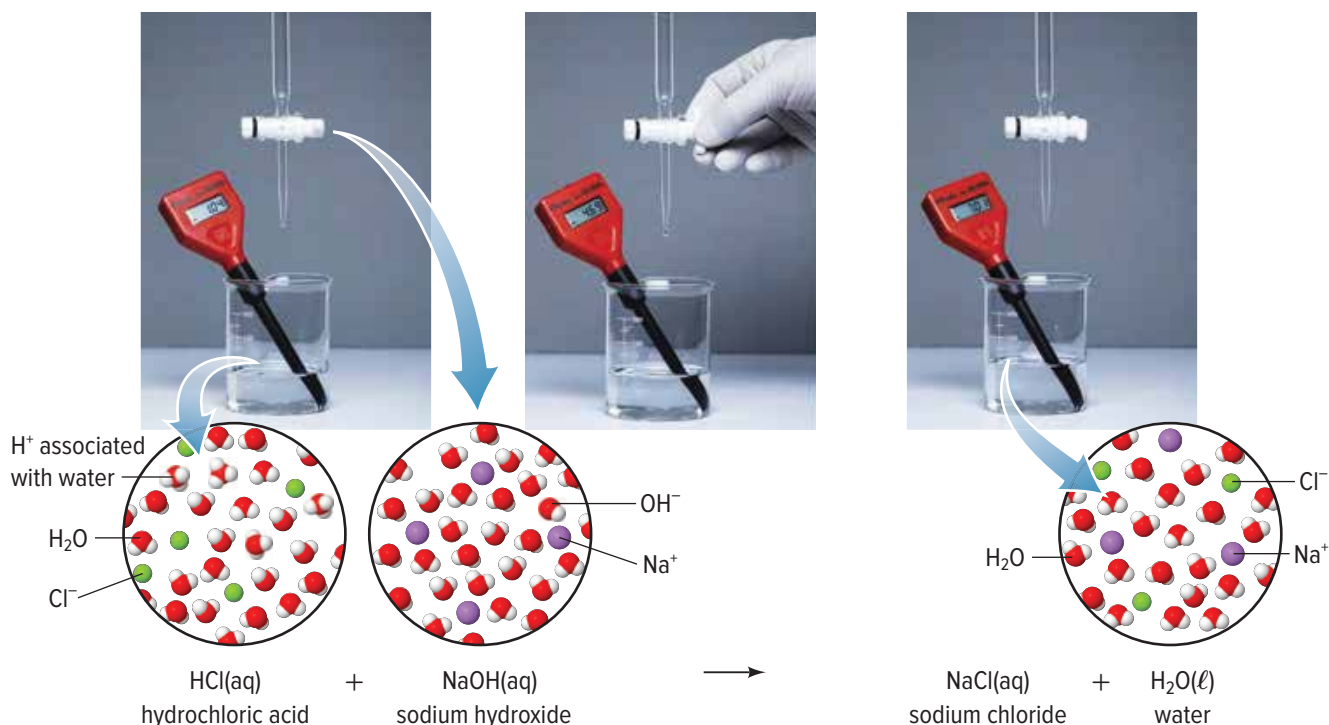
When an acid and a base are added together, it results in the removal of hydrogen ions and hydroxide ions from the solution, since their combination produces water. This causes the pH of the solution to approach 7.

Figure 2.41 shows a neutralization reaction. Notice how all the solutions are colourless. The progress of a neutralization reaction can be followed by measuring the pH of the solution as the acid and base are combined.

neutralization reaction a chemical reaction in which an acid reacts with a base to form a salt and water

Connect to Investigation 2-H on page 180

Figure 2.41 For all neutralization reactions, there is a net transfer of energy from the system to the surroundings. **Applying:** Are neutralization reactions exothermic or endothermic?



Neutralization Reactions in Our World

Neutralization is not just a chemical reaction that scientists carry out in the laboratory. It is one of the most common chemical reactions in everyday life. **Figure 2.42** shows a few examples of how we rely on reactions between acids and bases.



pH Adjustments

Monitoring pH is part of the regular maintenance of swimming pools and aquariums. Setting the correct pH relies on neutralization reactions so the water is safe to use.



Fishy Food

Fish odours are due to molecules that are bases. Lemon juice “neutralizes” the odour because the citric acid in the lemons reacts with the fishy-smelling base to produce water and a salt, which have no odour.



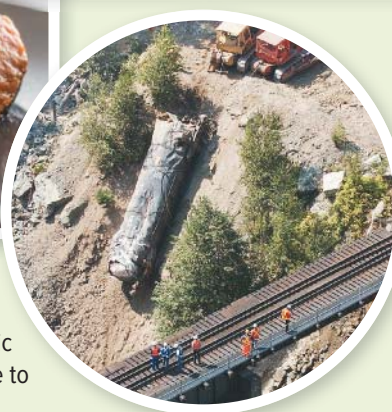
Antacids

The gastric pits in stomach lining secrete hydrochloric acid. Excess acid can cause a burning sensation in the stomach, chest, and throat. Antacids ease these symptoms, because they include basic hydroxide or carbonate compounds.



Acid Precipitation

Acid precipitation can harm crops and aquatic ecosystems. Adding crushed limestone (called liming) is used to raise the pH of an affected area. Crushed limestone is mostly calcium carbonate.



Chemical Spills

In 2005, a train derailment spilled 40 000 L of caustic soda (sodium hydroxide) into the Cheakamus River near Squamish, killing 500 000 fish. What is the condition of this river today, and who is responsible?

Connect to Investigation 2-I on page 183



Before you leave this page . . .

1. In your own words, describe what an acid and a base are.
2. Why is a neutralization reaction a type of double replacement reaction?
3. Write the predicted products and balance the chemical equation for the following neutralization reaction. You do not need to include the states of the products.

$$\text{HCl(aq)} + \text{Mg(OH)}_2\text{(aq)} \rightarrow$$

Figure 2.42

From medicinal products to our environment, neutralization reactions have many applications.

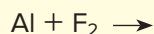
Identifying Reaction Types

Now that you have learned about the six types of reactions, use the following Sample and Practice Problems to help you identify the type of reaction that a chemical equation represents.

Sample Problem 1:

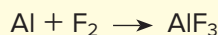
Predicting the Products of a Synthesis Reaction

Identify the type of reaction. Then predict the products and give the balanced chemical equation.

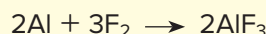


Solution

1. This reaction has two reactants and both are elements. Therefore, this is a synthesis reaction.
2. Since one reactant is a metal and the other reactant is a non-metal, the product will be a binary ionic compound composed of ions of both elements. Aluminum forms a Al^{3+} ion and fluorine forms a F^- ion. Therefore, they will produce the compound, AlF_3
3. Write the skeleton equation with the reactants and predicted product.



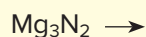
4. Write the balanced chemical equation.



Sample Problem 2:

Predicting the Products of a Decomposition Reaction

Identify the type of reaction. Then predict the products and give the balanced chemical equation.



Solution

1. This reaction has only one reactant. Therefore, this is a decomposition reaction.
2. The products will be the separate elements, Mg and N_2 .
3. Write the skeleton equation with the reactant and predicted products.



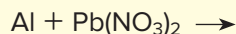
4. Write the balanced chemical equation.



Sample Problem 3:

Predicting the Products of a Replacement Reaction

Identify the type of reaction. Then predict the products and give the balanced chemical equation.



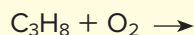
Solution

1. This reaction has two reactants; one is an element and the other is a compound. Therefore, this is a single replacement reaction.
2. Aluminum is a metal and it will replace lead(II) ions in the compound to form the compound $\text{Al}(\text{NO}_3)_3$. The element lead will be the other product.
3. Write the skeleton equation with the reactants and predicted products.
$$\text{Al} + \text{Pb}(\text{NO}_3)_2 \rightarrow \text{Pb} + \text{Al}(\text{NO}_3)_3$$
4. Write the balanced chemical equation.
$$2\text{Al} + 3\text{Pb}(\text{NO}_3)_2 \rightarrow 3\text{Pb} + 2\text{Al}(\text{NO}_3)_3$$

Sample Problem 4:

Predicting the Products of a Hydrocarbon Combustion Reaction

Identify the type of reaction. Then predict the products and give the balanced chemical equation.



1. This reaction has two reactants; one is oxygen and the other is a hydrocarbon, which is called propane. Therefore, this represents the combustion of a hydrocarbon.
2. The products of the combustion of a hydrocarbon are carbon dioxide and water.
3. Write the skeleton equation with the reactants and predicted products.
$$\text{C}_3\text{H}_8 + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$$
4. Write the balanced chemical equation. A common strategy for balancing these types of equations involves balancing the carbons first, then the hydrogens, and then the oxygens.
$$\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$$

Practice Problems

Identify the type of reaction. Then predict the products and give the balanced chemical equation.

1. $\text{Al} + \text{CuCl}_2 \rightarrow$
2. $\text{NaCl} + \text{AgNO}_3 \rightarrow$
3. $\text{C}_6\text{H}_{14} + \text{O}_2 \rightarrow$
4. $\text{Ca}(\text{OH})_2 + \text{HCl} \rightarrow$
5. $\text{KI} \rightarrow$
6. $\text{Zn} + \text{N}_2 \rightarrow$
7. $\text{Cd} + \text{Au}(\text{NO}_3)_3 \rightarrow$
8. $\text{Fe}_2\text{O}_3 \rightarrow$
9. $\text{K} + \text{Cl}_2 \rightarrow$
10. $\text{Cl}_2 + \text{CsBr} \rightarrow$

Check Your Understanding of Topic 2.4

OP Questioning and Predicting
 PC Planning and Conducting
 PA Processing and Analyzing
 E Evaluating
AI Applying and Innovating
 C Communicating

Understanding Key Ideas

- Synthesis reactions are also often referred to as combination reactions. Explain why this other name makes sense. PA
- Explain why most synthesis reactions are exothermic. What does this imply about the energy needed to break the reactant bonds compared to energy released when the product bonds form? Draw an energy level diagram that could represent a synthesis reaction. PA
- Develop a general equation that could represent a decomposition reaction. Why can you consider a decomposition reaction to be the reverse of a synthesis reaction? AI
- Copy and complete the following table in your notebook. PA C

	H ⁺ concentration greater than, less than, or equal to OH ⁻ concentration?	pH greater than, less than, or equal to 7?
Acid solution		
Base solution		
Neutral solution		

Connecting Ideas

- Use the following chemical equations to answer the questions below. PA C

$$\text{Mg(s)} + \text{AgNO}_3\text{(aq)} \rightarrow \text{Ag(s)} + \text{Mg(NO}_3)_2$$

$$\text{Na}_2\text{SO}_4\text{(aq)} + \text{BaCl}_2\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)} + \text{NaCl(aq)}$$
 - Identify each type of reaction.
 - Write the balanced chemical equation for each.

- For each skeleton equation below, classify the type of reaction as a synthesis, decomposition, single replacement, double replacement, neutralization, or combustion reaction. Justify your classification. Then write the balanced chemical equation for each. PA E
 - $\text{C}_4\text{H}_{10}\text{(g)} + \text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + \text{H}_2\text{O(g)}$
 - $\text{Li(s)} + \text{N}_2\text{(g)} \rightarrow \text{Li}_3\text{N(s)}$
 - $\text{AgNO}_3\text{(aq)} + \text{Na}_2\text{CrO}_4\text{(aq)} \rightarrow \text{NaNO}_3\text{(aq)} + \text{Ag}_2\text{CrO}_4\text{(s)}$
 - $\text{MgO(s)} \rightarrow \text{Mg(s)} + \text{O}_2\text{(g)}$
 - $\text{HI(aq)} + \text{KOH(aq)} \rightarrow \text{KI(aq)} + \text{H}_2\text{O(l)}$
 - $\text{AlPO}_4\text{(aq)} + \text{Mg(s)} \rightarrow \text{Al(s)} + \text{Mg}_3\text{(PO}_4)_2\text{(aq)}$
- For each of the following, identify the type of reaction that is represented. Then, predict the products and write the balanced chemical equations. You do not need to provide the states of the products. PA E
 - $\text{HCl(aq)} + \text{Mg(OH)}_2\text{(aq)} \rightarrow$
 - $\text{Al}_2\text{O}_3\text{(s)} \rightarrow$
 - $\text{Na(s)} + \text{Cl}_2\text{(g)} \rightarrow$
 - $\text{NH}_4\text{Cl(aq)} + \text{AgCH}_3\text{COO(aq)} \rightarrow$
 - $\text{NaI(aq)} + \text{Br}_2\text{(g)} \rightarrow$
 - $\text{Al(s)} + \text{CuCl}_2\text{(aq)} \rightarrow$

Making New Connections

- Society relies on hydrocarbon combustion.
 - What are the dangers associated with incomplete combustion of hydrocarbons?
 - Do you think hydrocarbons with a greater number of carbon and hydrogen atoms will produce more or less energy than smaller hydrocarbons? Explain.
 - Describe one advantage and one disadvantage of our use of hydrocarbon combustion. E C

What are the consequences of increasing ocean acidity?

What's the Issue?

Do you eat tuna, salmon, or other ocean fish? What about clams, crab, or lobster? Fish and other organisms from the ocean are the main source of protein for more than half of all the people in the world. However, our need for fossil fuels has created a problem that threatens the lives of these organisms, and thus the lives and livelihoods of people who depend on them.

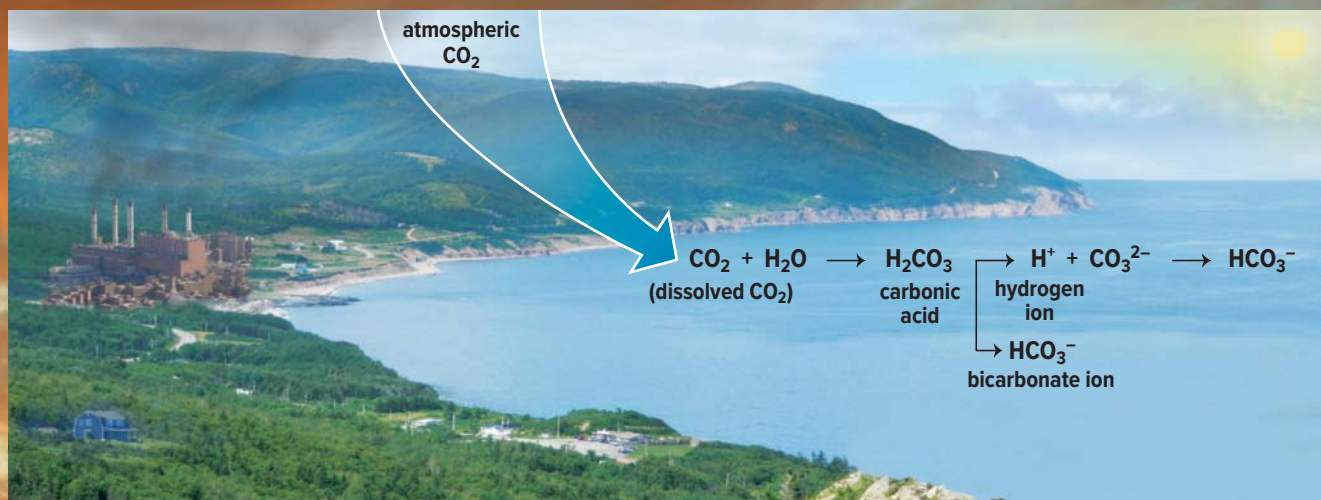
The Link Between Fossil Fuels and a More Acidic Ocean

When fossil fuels are burned for energy, they release carbon dioxide gas into the atmosphere. Carbon dioxide is a greenhouse gas and, as such, it plays an important role in the greenhouse effect that keeps our planet warm enough to sustain life. However, the concentration of carbon dioxide in the atmosphere has been increasing dramatically—mainly due to human activity. Carbon dioxide has been identified as the main contributor to global warming. But it also affects the chemistry of the ocean.

Interactions between the Ocean and the Atmosphere

Oceans are a huge, natural reservoir of carbon dioxide. At the surface, carbon dioxide in the atmosphere naturally dissolves in ocean water. This dissolved carbon dioxide sinks in masses to the deep ocean and returns to the surface hundreds of years later. If this natural uptake by the ocean did not occur, the accumulation of carbon dioxide in the atmosphere would be much greater.

When carbon dioxide gas dissolves in ocean water, it forms carbonic acid. Carbonic acid then produces hydrogen ions (H^+) and bicarbonate ions (HCO_3^-). Some of the hydrogen ions combine with naturally existing carbonate ions (CO_3^{2-}) to form additional bicarbonate ions. This means that with an increase in carbon dioxide in the atmosphere, the concentration of carbonate ions in the ocean decreases, and the concentration of hydrogen ions increases. An increase in the concentration of hydrogen ions causes the ocean water to become more acidic.



Consequences of Acidic Oceans

Many ocean organisms make their shells and skeletons from calcium carbonate, CaCO_3 . As the concentration of carbonate ions decreases, the ability of ocean organisms to make their shells and skeletons also decreases. Increased acidity of ocean water also causes the shells and skeletons of ocean organisms to become weaker. Some people have reported seeing holes in the shells of oysters and clams. As a result, the health and survival of these organisms are threatened.

A second problem involves microscopic plankton that form the base of food chains in many ocean ecosystems. If they cannot form their shells, they will die, their numbers will decrease, and food chains will be affected. Corals also use carbonate ions to build the skeletons of enormous reefs that support great numbers of organisms. Many of the fish and shellfish that people depend on live among coral reefs during their early years. As well, snails and other shelled organisms are food sources for other organisms—including people. Coral reefs also protect coastlines from damage during storms.



Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. What type of compound is calcium carbonate? What ions form when this compound is dissolved in water?
2. What is happening to the pH of our oceans? Explain your answer.
3. Brainstorm a list of possible solutions for the problems described in this feature. What changes could people make in their lives that could contribute to solutions?
4. What questions does this feature raise in your mind? In terms of further study, speculate on possible directions in which your questions could lead you.

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing
- Evaluating

Safety



- Wear safety goggles and gloves.
- The materials can become quite hot. Avoid touching the hot liquid or glass. Use a thermal glove or tongs.

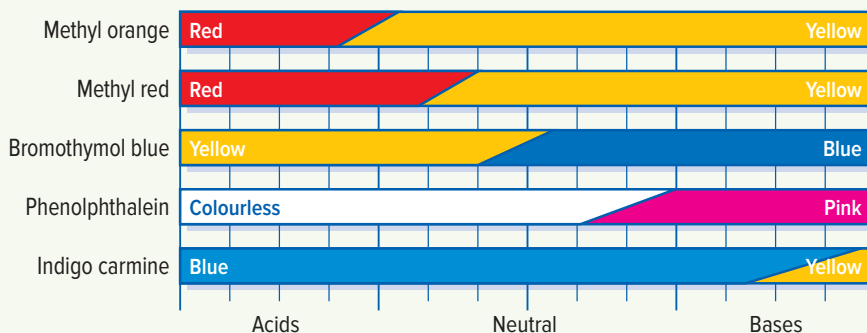
What You Need

- spot plates
- small test tubes
- masking tape
- unknowns 1, 2, 3, and 4
- Mg ribbon
- red and blue litmus paper
- bromothymol blue indicator
- methyl red indicator
- methyl orange indicator
- phenolphthalein indicator
- distilled water

Using Acid and Base Properties to Identify Four Unknowns

The following chart indicates the colour changes of some common acid-base indicators. The table below summarizes additional properties of acids and bases.

Indicator	pH Range in Which Colour Change Occurs	Colour Change as pH Increases
Methyl orange	3.2–4.4	red to yellow
Methyl red	4.8–6.0	red to yellow
Bromothymol blue	6.0–7.6	yellow to blue
Phenolphthalein	8.2–10.0	colourless to pink
Indigo carmine	11.2–13.0	blue to yellow



Property	Acid	Bases
Litmus test	turns blue litmus red	turns red litmus blue
Reaction with metal	yes (most)	no (most)
pH	less than 7	greater than 7

In this investigation, you will use the properties of four unknowns to help you identify them as acidic, basic, or neutral and determine the pH range of the solutions.

Question

How can the properties of a solution be used to identify it as acidic, basic, or neutral?

Procedure

1. Read through the information on the previous page and the list of chemicals provided. Your teacher may provide an additional list of available materials and associated information to use. Based on what is provided, design a procedure to identify each unknown as an acidic, basic, or neutral solution and to determine the pH or pH range of the solution. In designing your procedure, consider the following:
 - What tests will you perform and what information does each provide?
 - Will the tests provide all the information you need?
 - What materials do you need to perform the tests?
 - How much of the unknowns will you test? Are they provided as solids or liquids? What do you need to do if an unknown is a solid?
 - How will you keep track of the different unknown samples you are testing?
 - How will you record and organize the data you will be collecting?
2. As part of your procedure, include all safety precautions that must be taken, as well as clean-up and disposal instructions.
3. Have your teacher approve your procedure before proceeding.

Process and Analyze

1. Which unknowns were acidic and which were basic? Were any of the unknowns neutral? Provide evidence from your data that supports your answer.
2. What was the pH or pH range for each solution?

Conclude and Communicate

3. Arrange the unknowns in order of most acidic to least acidic.
4. Why might the pH range, and not the exact pH value, have been determined for a solution?
5. Why does using a universal indicator or pH meter provide more accurate results than acid-base indicators such as methyl red?

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing
- Evaluating

Safety



- Wear safety goggles, gloves, and a lab coat or apron.
- Handle chemicals safely. Avoid touching all reactants and products.
- Acid and base solutions are corrosive. Use them with care.
- When using a Bunsen burner, be sure you understand how to light and use it safely.
- Wash your hands thoroughly after completing this investigation.

What You Need

Part 1:

- Bunsen burner with matches or striker to light
- copper wire, 6 cm in length
- metal tongs

Part 2:

- 60 mL 3% hydrogen peroxide solution
- 250 mL beaker
- thermometer
- pinch of manganese dioxide, $\text{MnO}_2(\text{s})$
- stirring rod

Part 3:

- 50 mL copper(II) sulfate solution, 0.1 mol/L $\text{CuSO}_4(\text{aq})$
- 250 mL beaker
- thermometer
- shiny iron nail

Part 4:

- 20 mL calcium chloride solution, 0.1 mol/L $\text{CaCl}_2(\text{aq})$
- 250 mL beaker
- thermometer
- graduated cylinder
- 20 mL sodium carbonate solution, 0.1 mol/L $\text{Na}_2\text{CO}_3(\text{aq})$
- stirring rod

Part 5:

- tea light-style candle with base
- aluminum dish to collect wax drippings
- matches
- 100 mL beaker

Part 6:

- 0.1 mol/L $\text{HCl}(\text{aq})$
- 0.1 mol/L $\text{NaOH}(\text{aq})$
- graduated cylinder
- 100 mL beaker
- stirring rod
- pH paper (with universal indicator) and colour chart
- thermometer

Six Types of Reactions

In this investigation, you will carry out examples of different types of chemical reactions that you have studied. Use your understanding of reaction types to identify the types of reactions you perform.

Question

How can you use your knowledge of chemical reactions to classify reactions you carry out in the laboratory?

Procedure

Part 1: $\text{Cu(s)} + \text{O}_2 \rightarrow \text{CuO}$

1. Make a table for your observations.
2. Record the appearance of the copper wire.
3. Follow your teacher's instructions to light the Bunsen burner. Adjust the flame to obtain a blue flame with two distinct cones.
4. Use metal tongs to hold the middle of the wire in the hottest part of the flame (the top of the inner cone). Hold it there for about 1 min, then remove it. CAUTION: THE WIRE WILL BE VERY HOT. DO NOT TOUCH IT. HOLD IT WITH THE TONGS AS IT COOLS.
5. Record all observations while the wire is in the flame and after it is removed.
6. Clean up your work area and dispose of all materials as instructed by your teacher.

Part 2: $\text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{O} + \text{O}_2$

1. Make a table for your observations.
2. Add 60 mL of 3% hydrogen peroxide solution to the beaker.
3. Place a thermometer in the beaker and record the temperature of the solution.
4. Your teacher will add tiny amount of manganese dioxide into the hydrogen peroxide. Using a stirring rod, stir the solution. (The manganese dioxide is not a reactant. It helps make the reaction go faster.)
5. Record the temperature every 30 seconds as the reaction occurs.
6. Record your measurements and any other observations for about 10 minutes, until the reaction has slowed down or stopped.
7. Clean up your work area and dispose of all chemicals as instructed by your teacher.

Part 3:

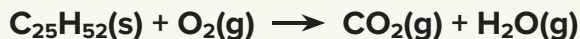
$\text{Fe(s)} + \text{CuSO}_4 \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu(s)}$

1. Make a table for your observations.
2. Place 50 mL of copper(II) sulfate solution into a beaker.
3. Carefully place a thermometer in the solution. Record the temperature and any other observations of this solution.
4. Place the iron nail into the solution in the beaker.
5. Record the temperature every 30 seconds and monitor what happens over several minutes. Record all your observations.
6. Put this reaction aside while you perform the remainder of the investigation. Then, record your final observations at the end of the investigation.

Part 4: $\text{CaCl}_2(\text{aq}) + \text{Na}_2\text{CO}_3(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{CaCO}_3(\text{s})$

1. Make a table for your observations.
2. Place 20 mL of calcium chloride solution into a small beaker. Using a thermometer, measure the temperature of the solution.
3. Rinse the graduated cylinder with water and then measure 20 mL of sodium carbonate solution. Pour this into the beaker containing calcium chloride.
4. Use a stirring rod to mix the solutions. Then record the temperature of the mixture.
5. Record the temperature every 30 seconds and monitor what happens over several minutes. Record all your observations.
6. Clean up your work area and dispose of all chemicals as instructed by your teacher.

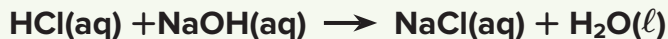
Part 5:



Candle wax is a mixture, but the formula $\text{C}_{25}\text{H}_{52}(\text{s})$ can be used to represent it.

1. Place the candle on the aluminum plate. Light the candle and place the beaker over it.
2. When the flame goes out, examine the inside of the beaker. Record your observations.
3. Clean up your work area and dispose of all materials as directed by your teacher.

Part 6:



Guided Inquiry

1. Using the materials provided, design a procedure for performing the neutralization reaction shown above. As part of your procedure include all safety precautions that must be taken, as well as proper clean-up and disposal that must be done.
2. Have your teacher approve your procedure before you begin.
3. Carry out your procedure and record all your observations.

Remember to go back to record your final observations of the reaction in Part 3. Then clean up your work area and dispose of any chemicals according to your teacher's instructions.

Process and Analyze

1. For each reaction, list the evidence for chemical change that you observed.
2. Based on your observations for the combustion reaction, did complete combustion or incomplete combustion occur? How do you know?

Conclude and Communicate

3. Write a word equation and a balanced chemical equation for each reaction. For Part 5, assume complete combustion.
4. Classify each of the reactions as a decomposition, synthesis, single replacement, double replacement, neutralization, or combustion reaction. Provide an explanation for each.
5. Where possible, identify the reaction as exothermic or endothermic. Provide an explanation for each.

Apply and Innovate

6. Hydrogen peroxide slowly decomposes over time, without the addition of any other chemical. What does that tell you about the hydrogen peroxide solution that can be purchased in the store? What is one potential safety hazard associated with this?

Skills and Strategies

- Planning and Conducting
- Processing and Analyzing
- Evaluating

Safety

- Safety goggles, a lab coat or apron, and gloves must be worn throughout the investigation.

What You Need

- 0.1 mol/L HCl(aq), to mimic stomach acid
- three different antacid tablets
- other materials, as determined by your procedure

Evaluate the Effectiveness of Antacids

Many kinds of antacids are available. In this investigation, you will design an experiment to determine which antacid is the best, based on the properties that you decided are the most important for an antacid.

Procedure

1. As a class, decide what properties make the best antacid. Different aspects to consider include:
 - How much is needed to neutralize a certain amount of acid?
 - What is the cost of the antacid?
 - Do some antacids have fewer side effects than others?
2. Working in small groups, design a procedure to determine the best antacid. Decide which antacids you will test, and make a list of equipment and materials you will need. Include all safety precautions, as well as how the materials will be disposed of.
3. Identify the variables for your experiment. There should be one independent variable (what you are changing from one trial to the next), and one dependent variable (what you are going to measure in each trial). All other factors in the experiment should be controlled (should stay the same in every trial).
4. Have your teacher approve your procedure and materials list.
5. Make a table to record your observations.
6. Carry out your procedure and record your results.
7. Clean up your work area and dispose of any chemicals according to your teacher's instructions.

Process and Analyze

1. What properties did the class decide were the most important in an antacid? Explain the rationale.
2. Write the balanced chemical equation for each neutralization reaction that occurs for each antacid.

Conclude and Communicate

3. Which antacid would you advise someone to use? Use your results to support your answer.